

plication (or front-end), you place a client (i.e., small amount of software) on a server in place of a traditional backup agent. This client eliminates duplicate backup data before sending it across the network. This bandwidth efficient approach is ideal for systems in remote offices, virtual environments (e.g., VMware ESX servers), or smaller offices with limited bandwidth. And unlike a traditional backup approach where backups streams flood the network, limiting the scheduling of backup jobs, client-side deduplication enables you to run many jobs simultaneously because a given backup job needs anywhere from 10-500 times less bandwidth.

Companies can leverage client-side deduplication for bandwidth constrained backup systems such as those found in remote offices and virtual environments to increase the reliability of recovery, to centralise backup data, and reduce storage costs.

Storage-based deduplication is gaining popularity in the datacenter as a way to help manage storage growth, recover data from disk, and expand disk-based disaster recovery to lower tiers of data. Unlike client-side deduplication, storage-based deduplication normally does not require any changes to your backup software. It can be deployed as a hardware storage appliance or as software solution that uses your own combination of servers and storage. Both approaches reduce the size of a backup after it has streamed across a network and through a backup server. Most customers have fewer concerns about backup bandwidth in the datacenter.

The storage efficiencies are the same as with client-side deduplication and typically reduce aggregate storage required for a backup (over a given retention time) by 10-20 times, when compared to tape. For data with short retention periods (e.g., one to four weeks), using deduplication can be especially appealing because it can improve the reliability of recovery and help companies control tape related costs in the face of growing data volumes.

With deduplication the traditional concept of a full backup, or a full copy, disappears. Instead every new backup (or copy) relies on the previous copy to ensure that only the unique blocks are stored. Of course, a full backup can be recovered at any time — even if you haven't written a typical full backup to disk or tape. Hardware redundancy (e.g., RAID 5) protects most systems, but customers need to remember that backups of their “dedupe storage system”, and not just replicate a copy, to add an important layer of protection.

How Can You Get Started With Deduplication?

Should client-side deduplication or storage-based deduplication replace all your current backup methods and media? Most likely not, because most environments require different degrees of

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protection and not all types of data are good candidates for deduplication. In other words, recovery point objectives (RPOs) and recovery time objectives (RTOs) should be matched to data protection methods. More recovery points might be better served by snapshots or continuous data protection (CDP). Faster recovery might be better served with snapshot methods or a SAN based backup to high-speed disk. And recovery requirements for data often change as the data ages. Finally, not all data types dedupe well, in particular compressed file formats used for music, photos, medicine, or research. For example, if a dedupe storage system cannot recover large amounts of data (or millions of small files) quickly, a user might instead choose to store backup data on high speed disk for the first week and then move this data to a dedupe storage target for the remainder of the retention time. The underlying principle when considering dedupe technology — evaluate the benefit and limitations of each approach alongside your RTO and RPO objectives.

Customers should deploy data deduplication based on specific needs within their environment. Controlling backup storage costs is an obvious imperative for many companies, but eliminating distributed tape-based backups may also present a cost-saving opportunity. Here are a few points to remember as you consider where to use this technology.

Assess where you can use this technology (e.g., client-side and/or storage-based) across all offices, datacenters, and systems — including virtual environments.

Evaluate how deduplication technology fits within your existing data protection environment (backup applications, storage, and servers).

Examine the trade-offs between storage appliances and software-based solutions for dedupe storage.

Determine whether you will need to export data out of your deduplication storage system to tape.

Remember to factor in the protection of your dedupe storage system (from data corruption) as part of your overall architecture.

Conclusion

According to some experts, data deduplication can reduce total backup storage usage by factors of 10:1 or more (depending on the nature of the data) when compared to traditional backup methods to tape. The bandwidth reductions delivered by client-side data deduplication technology are even more significant. When deployed as part of the overall backup strategy and used alongside other backup methods and media, there is no doubt that deduplication can bring both operational and economic benefits to companies. ■

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Has there been a change in the positioning of Red Hat in the market space post your acquisition of JBoss?

Initially, Red Hat focused on the Linux space but today we have a much broader value proposition in terms of our offerings to the enterprises. Virtualization, clustering and data management are some of the areas where we have made inroads.

The overall IT space has three layers that include hardware followed by the various business applications on top and finally the middle layer with various operating systems, database and the middleware.

The new and important market for us is in the solution stack which is the middleware space post our JBoss acquisition. It is in this layer that we see ourselves offering solutions to build IT infrastructure and drive business for companies.

What differentiates the JBoss enterprise SOA platform with other solutions available in the market?

JBoss Enterprise SOA Platform is an open source SOA product offering designed to help enterprises develop business solutions more rapidly and to improve productivity within and across enterprises.

The platform offers enterprises the opportunity to integrate applications and services, as well as streamline business processes in a

Open Source will become more mainstream

Gerry Messer, President, Asia Pacific and Japan, Red Hat

simple, open and cost-effective way as compared to expensive and monolithic proprietary SOA platforms. This next-generation SOA Platform coupled with the JBoss subscription can help add significant business and IT value to enterprise SOA integration projects.

Enterprises have already started using this platform for their new applications and we have also come across enterprises that used other middleware solutions migrating to this platform.

We have a clear strategy for this platform as to where it is today and where it would be in the next six to nine months. We are sure of making more improvements in this platform and maximise flexibility in a wide array of deployment scenarios from small-scale integration projects to enterprise-wide SOA integration.

What are the other features offered by this platform?

The platform also includes scalable clustering, Java Enterprise Edition technologies and a highly customisable base to meet ever-changing enterprise needs. The platform combines SOA, enterprise application integration (EAI), business process and rule management (BPM) and event-driven architecture (EDA) technologies to integrate services and applications and automate business processes for improved business productivity.

How do you see the growth of open source in the region?

The adoption rate in the region is strong and accelerating. Enterprises realise that using open source solutions, especially at the middle layer with open standards and access, helps to drive down costs of the IT infrastructure.

Open source will become more mainstream and this would mean that there would be a strong competition between the vendors. The major adopters of open source have been in the verticals of BFSI, telecom, government, education and healthcare. Nine out of ten global banks listed on the Wall Street are using Red Hat Linux to run their mission critical platforms.

Open source has stayed and proven itself. Enterprises no more discuss the issue whether open source is safe and secure and whether the software would perform. Gartner predicted that by 2011, 80 per cent of commercial software would be based on open source.

Comparing the open source adoption in India with other countries in APAC, we see that the government and the enterprise segments are looking at open source simultaneously. In other APAC countries it is either the government that has been the initial adopters of open source and then followed by enterprises or vice versa. ■